PERIPHERAL TROUBLESHOOTING

Practical Assignment

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Topic	Webcam and Printer Setup, Testing, and Troubleshooting
Practical Focus	Windows computer peripherals

This assignment explains how to connect and test a USB webcam, install and test a printer, diagnose a printer that is not detected, and fix a webcam that works in the Windows Camera app but not in Zoom.

Introduction

Computer peripherals are devices connected to a computer to provide extra functions. A webcam captures video and photos, while a printer produces paper copies of documents and images. For these devices to work properly, the physical connection, Windows settings, device drivers, and application permissions must all be correct.

Troubleshooting means checking a problem in an organised way, starting with simple causes before moving to more advanced settings. This assignment uses safe, practical checks that can be followed on a Windows computer. The exact menu names may look slightly different depending on the Windows version and device model.

Objectives

• Connect a USB webcam and confirm that it can capture a photo.
• Install a printer and print a Windows test page.
• Use Device Manager and Scan for hardware changes to recover a disconnected USB printer.
• Diagnose why a webcam works in Camera but not in Zoom.
• Understand common causes, solutions, and prevention tips.

Safety and preparation

• Keep the computer and peripherals on a stable, dry surface.
• Do not force USB plugs into ports; check their orientation and insert gently.
• Download drivers only from Windows Update or the official manufacturer website.
• Avoid unknown driver-download websites and unnecessary driver-update utilities.
• Before changing settings, close apps that may be using the camera or printer.

Practical 1: Connect and Test a USB Webcam

Aim

To connect a USB webcam, allow Windows to detect it, install any required driver, and verify its operation using the built-in Camera app.

Requirements

A Windows computer, a USB webcam, an available USB port, and the Windows Camera app. Internet access may be needed if Windows must download a driver.

Procedure

1. Inspect the webcam. Check the cable and USB connector for visible damage. If the webcam has a privacy shutter or lens cover, open it. Place the camera securely on the monitor or desk.

2. Connect the webcam. Plug it directly into a USB port on the computer. If possible, test a port on the computer itself rather than a USB hub. Wait a short time while Windows detects the device.

3. Allow driver installation. Windows often installs a basic compatible driver automatically. A notification may appear while the device is being set up. Keep the computer connected and wait for setup to finish. If Windows does not recognise the webcam, open **Settings** → **Windows Update** and check for updates, including available optional driver updates when appropriate.

4. Check Device Manager. Right-click Start and select **Device Manager**. Look under categories such as **Cameras**, **Imaging devices**, or **Sound, video and game controllers**. The exact category depends on the webcam. If the device appears without a warning icon, Windows is detecting it. If it appears under Unknown devices or has a warning symbol, check its properties and driver status.

5. Open Camera. Open the Start menu, search for **Camera**, and launch the Windows Camera app. If prompted, allow camera access. Select the correct camera if the app offers more than one option.

6. Verify the image. Point the webcam toward a well-lit subject. Confirm that a live preview appears and that the image updates when the camera moves. Press the photo capture button to take a test photo. Open the Camera app's photo preview or the saved image location to confirm that the photo was captured.

7. Record the result. Note the webcam name if visible, the USB port used, whether Windows installed a driver automatically, and whether the live preview and photo capture worked.

If the webcam does not work

• Try another USB port and reconnect the camera.
• Restart the computer and test the webcam again.
• Close other video-call or recording apps that may be using the camera.
• In **Settings** → **Privacy & security** → **Camera** (wording can vary), turn on camera access and allow desktop apps to access the camera.
• In Device Manager, right-click the webcam and choose **Update driver**. If the device still fails, check the manufacturer's official support page for the correct model-specific driver.
• Test the webcam on another compatible computer if available. This helps determine whether the issue is with the webcam or the original computer.

Expected result

> The webcam is detected by Windows, appears in the Camera app, displays a live image, and successfully captures a test photo.

Practical 2: Set Up a Local Printer and Print a Test Page

Aim

To connect a USB or network printer, install the correct driver, and print a test page to confirm that Windows can communicate with the printer.

Requirements

A printer, suitable paper, power cable, USB cable for a USB printer or access to the same network for a network printer, and the correct printer driver if Windows does not install one automatically.

Procedure for a USB printer

1. Prepare the printer. Place the printer on a stable surface. Connect its power cable and turn it on. Load paper in the tray and check that there is ink or toner. Clear any paper-jam or low-supply warning before testing.

2. Connect the USB cable. Connect the printer's USB cable to the printer and computer. Follow the printer manufacturer's instructions about when to connect the cable; many modern printers can be connected while Windows is running.

3. Add the printer in Windows. Open **Settings** → **Bluetooth & devices** → **Printers & scanners** in Windows 11, or the corresponding Printers & scanners page in Windows 10. Select **Add device** or **Add a printer or scanner**. Wait for Windows to search, then select the correct printer if it appears.

4. Install the correct driver if needed. Windows may install a compatible driver automatically. If the printer is not found, or features are missing, identify the exact printer model from its label or display and visit the official printer manufacturer's support website. Choose the driver that matches the model and Windows version. Follow the installation instructions; do not install a driver for a similar but different model.

5. Set it as the default printer if required. In Printers & scanners, open the printer's settings and choose the default option if you want Windows to use it automatically. This is optional and is not required just to print a test page.

6. Print a test page. Open the printer's page in Printers & scanners. Select **Printer properties** where available, then choose **Print Test Page**. Depending on the driver, the option may be under a related printer-management

menu. Wait for the printer to receive the job and print the page.

7. Check the output. Confirm that a page comes out, the text and test pattern are visible, and the printer does not display an error. If the page is blank or poor quality, check ink or toner, paper placement, and the printer's maintenance options.

Procedure for a network printer

A network printer may connect through Wi-Fi or Ethernet. Make sure the printer is connected to the intended network and that the computer is connected to the same network or an authorised network that can reach the printer. Open Printers & scanners, choose Add device, and select the printer when it appears. If it does not appear, use the manufacturer's approved setup method or add it by its network address if your administrator or printer documentation provides that information. Install the correct driver and print a test page.

If the test page does not print

- Check that the printer is powered on and has no warning lights or display errors.
- Check paper, ink or toner, and any paper jam.
- Verify the USB cable is firmly connected, or verify the network connection for a network printer.
- In the print queue, cancel stuck jobs and confirm the printer is not paused or set to offline.
- Confirm that the correct printer is selected and that its driver matches the model.
- Restart the printer and computer, then try printing the test page again.

Expected result

Windows lists the correct printer, the driver is installed, and the printer successfully prints a readable test page.

Practical 3: Fix a Printer Not Detected After Unplugging USB

Aim

To simulate a USB printer connection problem, identify the missing or affected device in Device Manager, reconnect the cable, and ask Windows to scan for hardware changes.

Understanding the issue

When the USB cable is unplugged, the computer loses its physical connection to the printer. Windows may temporarily remove the printer from the active device list or show it as unavailable. This does not automatically mean that the printer driver is damaged. The first check should be the cable and USB port.

Procedure

1. Confirm normal operation first. With the printer connected, check that it appears under Printers & scanners and, if possible, print a test page. This gives you a baseline to compare with the simulated problem.

2. Simulate the problem. Leave the printer powered on and carefully unplug the USB cable from the computer. Do not pull the cable by its wire; hold the connector. The printer may remain listed in Windows, but it may show as unavailable or fail to accept new print jobs.

3. Open Device Manager. Right-click the Start button and select Device Manager. Inspect categories such as **Printers**, **Print queues**, **Universal Serial Bus controllers**, and **Other devices**. Category names depend on the Windows version and printer driver.

4. Look for clues. Check for a yellow warning icon, an Unknown device entry, or the absence of the printer while the cable is disconnected. A warning icon can indicate a device or driver issue, but it is not proof of a single cause. Open the device's Properties and read the Device status message if an entry is present.

5. Reconnect the USB cable. Plug the cable firmly back into the printer and computer. If the device is still not detected, try a different USB port. If a second compatible USB cable is available, test it as well. Wait for Windows to react to the reconnection.

6. Scan for hardware changes. In Device Manager, select the computer name at the top of the device tree, open the **Action** menu, and select **Scan for hardware changes**. Windows will search for connected hardware and may rediscover the printer. If the Action menu is not visible, maximise the window or check the menu bar.

7. Verify the printer. Return to Settings → Printers & scanners. Check whether the printer is listed and ready. Open the print queue to ensure it is not paused or offline, then print a test page.

8. If it remains undetected. Restart the printer and computer. Try a different USB port and cable, check the printer manufacturer's setup instructions, and install or repair the official driver if necessary. If Device Manager reports a specific error code, use that message to guide the next step rather than guessing.

Troubleshooting decision table

Observation	Possible cause	Action
Printer absent after unplugging	Normal result of losing USB connection	Reconnect cable and scan for hardware changes
Printer absent after reconnecting	Loose cable, faulty port/cable, or detection issue	Reseat cable; try another port or cable
Unknown device or warning icon	Driver or device communication problem	Read Properties → Device status; update/install official driver if needed
Printer listed but offline	Queue/status issue or printer unavailable	Check power/network/USB, queue status, and clear stuck jobs

Observation	Possible cause	Action
Printer ready but no page prints	Paper/supply issue or stuck job	Check display, paper, ink/toner, queue, then test again

Important note

Scan for hardware changes can help Windows rediscover a connected device, but it cannot repair a broken cable or a printer that is switched off. Always check the physical connection first. Do not uninstall devices or remove drivers unless simpler checks fail and you understand which printer entry is being changed.

Expected result

After reconnecting the USB cable and scanning for hardware changes, Windows detects the printer again and a test page can be printed.

Practical 4: Webcam Works in Camera but Not in Zoom

Aim

To identify and fix common software, permission, and device-selection problems when a webcam works in the Windows Camera app but does not display video in Zoom.

Why this happens

If the webcam works in Camera, the hardware and basic Windows driver are probably functioning. The problem may be specific to Zoom: another app may be using the camera, Zoom may not have permission, the wrong camera may be selected, or the Zoom app may need to be restarted or updated.

Procedure

1. Close the Camera app. Close Camera and any other app that might use video, such as a browser video-call tab, Teams, or another meeting app. Some webcams or drivers do not allow two apps to use the camera at the same time. Wait a few seconds, then open Zoom again.

2. Check Windows camera permissions. Open Settings and go to **Privacy & security** → **Camera** on Windows 11, or **Privacy** → **Camera** on Windows 10. Turn on camera access for the device and allow apps to access the camera. Also enable the setting that allows desktop apps to access the camera, because Zoom desktop is a desktop application. Names and layout can vary by Windows version.

3. Select the correct camera in Zoom. Open Zoom and enter its Settings. Go to the Video or Video & effects section, depending on the version. Find the Camera dropdown and choose the USB webcam rather than a virtual camera, built-in camera, or unrelated device. Look for a preview image.

4. Check the physical webcam. Make sure the USB cable is secure and the lens cover or privacy shutter is open. If the webcam has a physical privacy switch, turn it off. If the image remains blank, reconnect the camera and wait for Windows to detect it.

5. Check Zoom's in-meeting controls. During a meeting, find the Start Video or Stop Video control. If video is stopped, select Start Video. If Zoom asks for permission, allow camera access. If the host or organisation has restrictions, follow the relevant meeting or administrator instructions.

6. Restart Zoom. Leave the meeting if appropriate and fully close Zoom, including any running background instance if visible. Reopen it and test the video preview. If necessary, restart the computer to release a camera session that may be stuck.

7. Update Zoom and Windows. Use Zoom's built-in update option or the official Zoom download page to obtain a supported version. Check Windows Update for available updates. Avoid downloading software from unofficial sites.

8. Check Device Manager if Zoom still has no video. Open Device Manager and confirm that the webcam appears without a warning icon. If an error is shown, open Properties and read Device status. Use Update driver or the webcam manufacturer's official support page if a driver repair is needed. Since Camera already works, avoid changing drivers unless there is evidence of a driver issue.

9. Test again. Open Zoom Settings → Video and check the preview. Then start or join a test meeting if available and confirm that other participants can see the video. If possible, test the webcam in another trusted video-call application to see whether the issue is limited to Zoom.

Quick troubleshooting checklist

- Close Camera and other apps using the webcam.
- Confirm Windows camera access and desktop-app camera access are enabled.
- Select the correct USB webcam in Zoom's video settings.
- Check the USB cable and lens privacy cover.
- Restart Zoom and, if needed, restart the computer.
- Update Zoom from its official source and check Windows updates.
- Inspect Device Manager for a warning icon or error message.
- Test the preview again and record the final result.

Expected result

Zoom displays the same webcam's live video that appears in the Windows Camera app, and the camera works during a Zoom meeting.

Results and Observation Record

Use this table to record what happened during the practical session. Fill in the result after performing each activity on the actual computer; do not mark a task successful unless you tested it.

Practical task	What to verify	Result / notes
USB webcam setup	Detected in Windows; live preview works; photo captured	_____
Printer installation	Correct printer listed; test page printed	_____
Printer USB reconnection	Device Manager scan detects printer again	_____
Zoom webcam troubleshooting	Camera preview works in Zoom and in a meeting	_____

Common troubleshooting habits

- Start with the simplest explanation: power, cable, USB port, paper, and privacy shutter.
- Change one thing at a time so you can tell which action fixed the problem.
- Read error messages carefully instead of assuming that every issue is a driver problem.
- Restart the app after changing its permissions or camera selection.
- Use Device Manager to inspect device status, but remember that the physical connection must be working first.
- Use official manufacturer support pages for drivers and instructions.
- After every fix, repeat the original test to confirm that the problem is actually solved.

Conclusion

In this practical assignment, the main steps for setting up and troubleshooting a USB webcam and printer were explained. A webcam should be detected by Windows and tested in the Camera app. A printer should be installed with the correct driver and confirmed by printing a test page. If a USB printer stops being detected after its cable is unplugged, reconnecting the cable and selecting Scan for hardware changes can help Windows detect it again. If a webcam works in Camera but not in Zoom, checking camera permissions, closing other camera apps, selecting the correct device, and restarting Zoom are useful first solutions.

The most important lesson is to troubleshoot in a clear order: check the physical connection, verify Windows detection, check settings and permissions, update software only when needed, and test again after making a change.

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